

RESEARCH INTERESTS

Robot learning, computer vision, and embodied AI

Grounding reasoning in robotic manipulation and developing general-purpose robotic foundation models that reason before they act.

CURRENT APPOINTMENTS

- 2026–present **Presidential Young Professor (Tenure-track), National University of Singapore**
School of Computing, Singapore; Director, MAGIC Lab
- 2026–present **Research Scientist, A*STAR IAIC (Joint Appointment)**
Singapore

EDUCATION

- 2022–2026 **Ph.D. in Computer Science and Engineering, University of Washington**
Seattle, USA — Advisors: Dieter Fox and Ranjay Krishna
- 2022–2024 **M.S. in Computer Science and Engineering, University of Washington**
Seattle, USA — CGPA: 3.91/4.00 (Magna Cum Laude)
- 2018–2021 **B.Eng. in Electrical and Electronic Engineering, Nanyang Technological University**
Singapore — CGPA: 4.75/5.00 (Highest Distinction)
- 2020 **Exchange Student, Georgia Institute of Technology**
School of Electrical and Computer Engineering, Atlanta, GA, USA
- 2013–2016 **Diploma with Merit in Engineering Science, Ngee Ann Polytechnic**
Singapore — CGPA: 3.97/4.00

RESEARCH PUBLICATIONS

* Equal contribution; † Second author; ‡ Equal advising; § Core contributor. Jiafei Duan is shown in bold.

Full publication list: jiafei1224.github.io

- Gehao Zhang, Weikai Huang, Shailesh Shailesh, Yiyang Peng, **Jiafei Duan**‡, Ranjay Krishna‡. “Grounded Action Model: 3D Grounding as a Foundation for Robotics.” *arXiv*, 2026.
- Jianman Lin*, Shailesh Shailesh*, Zhongyi Luo, **Jiafei Duan**. “Breaking the Vision–Action Shortcut: Latent Interface Training for Generalizable Robot Foundation Models.” *Preprint*, 2026.
- Haoquan Fang*, **Jiafei Duan***, Donovan Clay§, Sam Wang§, Shuo Liu§, Weikai Huang§, Xiang Fan§, Wei-Chuan Tsai§, Shirui Chen§, Yi Ru Wang§, Shanli Xing§, Jaemin Cho, Jae Sung Park, Ainaz Eftekhari, Peter Sushko, Karen Farley, Angad Wadhwa, Cole Harrison, Winson Han, Ying-Chun Lee, Eli Vanderbilt, Rose Hendrix, Suveen Ellawela, Lucas Ngoo, Joyce Chai, Zhongzheng Ren, Ali Farhadi, Dieter Fox, Ranjay Krishna. “MolmoAct2: Action Reasoning Models for Real-World Deployment.” *CoRL*, 2026 — *Spotlight (top 4.4% of submissions)*.
- Weikai Huang*, Jieyu Zhang*, Sijun Li, Taoyang Jia, **Jiafei Duan**, Yunqian Cheng, Jaemin Cho, Matthew Wallingford, Rustin Sotk, Chris Dongjoo Kim, Shuo Liu, Donovan Clay, Taira Anderson, Winson Han, Ali Farhadi, Bharath Hariharan, Zhongzheng Ren‡, Ranjay Krishna‡. “WildDet3D: Scaling Promptable 3D Detection in the Wild.” *arXiv*, 2026.
- Abhay Deshpande*, Maya Guru*, Rose Hendrix*, Snehal Jauhari*, Ainaz Eftekhari, Rohan Tripathi, Max Argus, Jordi Salvador, Haoquan Fang, Matthew Wallingford, Wilbert Pumacay, Yejin Kim, Quinn Pfeifer, Ying-Chun Lee, Piper Wolters, Omar Rayyan, Mingtong Zhang, **Jiafei Duan**, Karen Farley, Winson Han, Eli Vanderbilt, Dieter Fox, Ali Farhadi, Georgia Chalvatzaki, Dhruv Shah, Ranjay Krishna. “MolmoB0T: Large-Scale Simulation Enables Zero-Shot Manipulation.” *CoRL*, 2026 — *Oral (top 1.0% of submissions)*.
- Yalcin Tur, Jalal Naghiyev, Haoquan Fang, Wei-Chuan Tsai, **Jiafei Duan**‡, Dieter Fox‡, Ranjay Krishna‡. “Recurrent-Depth VLA: Implicit Test-Time Compute Scaling of Vision-Language-Action Models via Latent Iterative Reasoning.” *arXiv*, 2026.
- Shirui Chen, Cole Harrison, Ying-Chun Lee, Angela Jin Yang, Jason Ren, Lillian J. Ratliff, **Jiafei Duan**‡, Dieter Fox‡, Ranjay Krishna‡. “TOPReward: Token Probabilities as Hidden Zero-Shot Rewards for Robotics.” *CoRL*, 2026.

- Jianing Zhang*, Chenhao Zheng*, Yajun Yang, Max Argus, Rustin Soraki, Winson Han, Taira Anderson, Chun-Liang Li, Shuo Liu, **Jiafei Duan**, Zhongzheng Ren, Jieyu Zhang, Ranjay Krishna. "MolmoMotion: Forecasting Point Trajectories in 3D with Language Instruction." *NeurIPS, 2026 — Spotlight (1% acceptance rate)*.
- Shuo Liu, Ishneet Sukhvinder Singh, Yiqing Xu, **Jiafei Duan**†, Ranjay Krishna†. "VLS: Steering Pretrained Robot Policies via Vision-Language Models." *CoRL, 2026*.
- Zijun Lin, **Jiafei Duan**, Haoquan Fang, Dieter Fox†, Ranjay Krishna†, Cheston Tan, Bihan Wen. "FailSafe: Reasoning and Recovery from Failures in Vision-Language-Action Models." *IROS, 2026*.
- Yejin Kim*, Wilbert Pumacay*, Omar Rayyan*, Max Argus*, Winson Han, Eli VanderBilt, Jordi Salvador, Abhay Deshpande, Rose Hendrix, Snehal Jauhri, Shuo Liu, Nur Muhammad Mahi Shafiullah, Maya Guru, Ainaz Eftekhari, Karen Farley, Donovan Clay, **Jiafei Duan**, Arjun Guru, Piper Wolters, Alvaro Herrasti, Ying-Chun Lee, Georgia Chalvatzaki, Yuchen Cui, Ali Farhadi, Dieter Fox, Ranjay Krishna. "MolmoSpaces: A Large-Scale Open Ecosystem for Robot Navigation and Manipulation." *RSS, 2026 — Oral*.
- Suvir Mirchandani*, Mia Tang*, **Jiafei Duan**, Jubayer Ibn Hamid, Michael Cho, Dorsa Sadigh. "RoboCade: Gamifying Robot Data Collection." *ICRA 2026*.
- Jason Lee*§, **Jiafei Duan***§, Haoquan Fang*§, Yuquan Deng§, Shuo Liu§, Boyang Li, Bohan Fang, Jieyu Zhang, Yi Ru Wang§, Sangho Lee, Winson Han, Wilbert Pumacay, Angelica Wu, Rose Hendrix, Karen Farley, Eli VanderBilt, Ali Farhadi, Dieter Fox§†, Ranjay Krishna§†. "MolmoAct: Action Reasoning Models that Reason in Space." *ICRA, 2026*.
- Yi Ru Wang, Carter Ung, Grant Tannert, **Jiafei Duan**, Josephine Li, Amy Le, Rishabh Oswal, Markus Grotz, Wilbert Pumacay, Yuquan Deng, Ranjay Krishna, Dieter Fox, Siddhartha Srinivasa. "RoboEval: Where Robotic Manipulation Meets Structured and Scalable Evaluation." *ICRA 2026*.
- Ainaz Eftekhari, Rose Hendrix, Luca Weihs, **Jiafei Duan**, Ege Caglar, Jordi Salvador, Alvaro Herrasti, Winson Han, Eli VanderBilt, Aniruddha Kembhavi, Ali Farhadi, Ranjay Krishna, Kiana Ehsani, Kuo-Hao Zeng. "The One RING: A Robotic Indoor Navigation Generalist." *ICRA, 2026 — Oral*.
- Som Sagar, **Jiafei Duan**, Sreevishakh Vasudevan, Yifan Zhou, Heni Ben Amor, Dieter Fox, Ransalu Senanayake. "RoboMD: Uncovering Robot Vulnerabilities through Semantic Potential Fields." *ICLR, 2026*.
- Abhay Deshpande, Yuquan Deng, Arijit Ray, Jordi Salvador, Winson Han, **Jiafei Duan**, Kuo-Hao Zeng, Yuke Zhu, Ranjay Krishna, Rose Hendrix. "GraspMolmo: Generalizable Task-Oriented Grasping via Large-Scale Synthetic Data Generation." *CoRL 2025*.
- Long Cheng*, **Jiafei Duan***, Yi Ru Wang†, Ranjay Krishna†, Dieter Fox†. "PointArena: Probing Multimodal Grounding Through Language-Guided Pointing." *arXiv, 2025*.
- Arijit Ray, **Jiafei Duan**, Reuben Tan, Dina Bashkirova, Rose Hendrix, Kiana Ehsani, Aniruddha Kembhavi, Bryan A. Plummer, Ranjay Krishna, Kuo-Hao Zeng, Kate Saenko. "SAT: Spatial Aptitude Training for Multimodal Language Models." *COLM 2025*.
- Haoquan Fang, Markus Grotz, Wilbert Pumacay, Yi Ru Wang, Dieter Fox†, Ranjay Krishna†, **Jiafei Duan**†. "SAM2Act: Integrating Visual Foundation Model with A Memory Architecture for Robotic Manipulation." *ICML, 2025*.
- **Jiafei Duan**, Wilbert Pumacay, Nishanth Kumar, Yi Ru Wang, Shulin Tian, Wentao Yuan, Ranjay Krishna, Dieter Fox, Ajay Mandlekar†, Yijie Guo†. "AHA: A Vision-Language-Model for Detecting and Reasoning Over Failures in Robotic Manipulation." *ICLR, 2025*.
- **Jiafei Duan**, Wentao Yuan, Wilbert Pumacay, Yi Ru Wang, Kiana Ehsani, Dieter Fox, Ranjay Krishna. "Manipulate-Anything: Automating Real-World Robots using Vision-Language Models." *CoRL 2024*.
- Wentao Yuan, **Jiafei Duan**, Valts Blukis, Wilbert Pumacay, Ranjay Krishna, Adithyavairavan Murali, Arsalan Mousavian, Dieter Fox. "RoboPoint: A Vision-Language Model for Spatial Affordance Prediction for Robotics." *CoRL 2024*.
- Jun Wang, Chun-Cheng Chang*, **Jiafei Duan***, Dieter Fox, Ranjay Krishna. "EVE: Enabling Anyone to Train Robots Using Augmented Reality." *UIST 2024*.
- Ainaz Eftekhari*, Kuo-Hao Zeng*, **Jiafei Duan**, Ali Farhadi, Ani Kembhavi, Ranjay Krishna. "Selective Visual Representations Improve Convergence and Generalization for Embodied AI." *ICLR 2024 (Spotlight)*.
- Wilbert Pumacay*, Ishika Singh*, **Jiafei Duan***, Ranjay Krishna, Jesse Thomason, Dieter Fox. "THE COLOSSEUM: A Benchmark for Evaluating Generalization for Robotic Manipulation." *RSS, 2024 — Oral*.
- Samson Yu, Kelvin Lin, Anxing Xiao, **Jiafei Duan**, Harold Soh. "Octopi: Object Property Reasoning with Large Tactile-Language Models." *RSS, 2024 — Oral*.
- Yi Ru Wang, **Jiafei Duan**, Dieter Fox, Siddhartha Srinivasa. "NEWTON: Are Large Language Models Capable of Physical Reasoning?." *Findings of EMNLP 2023*.
- **Jiafei Duan**, Yi Ru Wang, Mohit Shridhar, Dieter Fox, Ranjay Krishna. "AR2-D2: Training a Robot without a Robot." *Conference on Robot Learning (CoRL) 2023*.
- Arijit Dasgupta, **Jiafei Duan**, Marcelo H. Ang Jr, Yi Lin, Su-hua Wang, Renee Baillargeon, Cheston Tan. "A Benchmark for Modeling Violation-of-Expectation in Physical Reasoning Across Event Categories." *Cogsci 2023*.
- Jenny Zhang, Samson Yu, **Jiafei Duan**, Cheston Tan. "Good Time to Ask: A Learning Framework for Asking for Help in Embodied Visual Navigation." *Ubiquitous Robot (UR) 2023, Best Paper Award*.
- **Jiafei Duan***, Samson Yu*, Soujanya Poria, Bihan Wen, Cheston Tan. "PIP: Physical Interaction Prediction via Mental Simulation with Span Selection." *ECCV, 2022*.
- **Jiafei Duan***, Arijit Dasgupta*, Jason Fischer, Cheston Tan. "A Survey on Machine Learning Approaches for Modelling Intuitive Physics." *IJCAI, 2022 — Survey Track, Oral*.
- **Jiafei Duan**, Samson Yu, Hui Li Tan, Hongyuan Zhu, Cheston Tan. "A Survey of Embodied AI: From Simulators to Research Tasks." *IEEE Transactions on Emerging Topics in Computational Intelligence (TETCI), 2021*.
- **Jiafei Duan**, Samson Yu, Hui Li Tan, Cheston Tan. "ActioNet: An Interactive End-to-End Platform for Task-Based Data Collection and Augmentation in 3D Environments." *ICIP 2020*.

NON-ARCHIVAL PUBLICATIONS

- Jieyi Ye, **Jiafei Duan**, Samson Yu, Bihan Wen, Cheston Tan. "ABCDE: An Agent-Based Cognitive Development Environment." *Computer Vision and Pattern Recognition Embodied AI Workshop (CVPRW)* 2022.
- Arijit Dasgupta, **Jiafei Duan**, Marcelo H. Ang Jr, Cheston Tan. "AVoE: A Synthetic 3D Dataset on Understanding Violation of Expectation for Artificial Cognition." *Conference on Neural Information Processing Systems (NeurIPS) 2021, Physical Reasoning and Inductive Biases for the Real World Workshop*.
- **Jiafei Duan**, Samson Yu, Cheston Tan. "SPACE: A Simulator for Physical Interactions and Causal Learning in 3D Environments." *IEEE International Conference for Computer Vision (ICCV) 2021, 1st Simulation Technology for Embodied AI Workshop*.

GRANTS

- **2026**: National Robotics Programme (NRP) 2.0 Funding Initiative — "A Singapore Dataset and Benchmark for Embodied AI", **S\$2,000,000**.
- **2026**: Cortex AI Gift Grant, **S\$24,000**.

RESEARCH EXPERIENCE

- 2023–2026 **Graduate Student Researcher, Allen Institute for AI**
PRIOR and Robotics teams; mentors: Ranjay Krishna and Dieter Fox
- 2025–2026 **Research Scientist Intern, Allen Institute for AI**
Advisor: Ranjay Krishna
- 2024 **Research Scientist Intern, NVIDIA Seattle Robotics Lab**
Advisors: Ajay Mandlekar and Yijie Guo
- 2021–2022 **AI Research Engineer, Institute for Infocomm Research, A*STAR**
Supervisor: Cheston Tan Yin Chet
- 2020–2021 **Final Year Undergraduate Researcher, Nanyang Technological University**
Advisor: Bihan Wen
- 2020 **Visiting Research Scholar, Georgia Institute of Technology**
Advisor: Ayanna Howard

TEACHING

- 2026 **Instructor, NUS CS6283: Robot Learning in the Era of Foundation Models**
- Winter 2026 **Co-instructor, UW CSE 571: Robotics**
With Dieter Fox
- Spring 2025 **Teaching Assistant, UW CSE 571: Robotics**
- Winter 2023 **Head Teaching Assistant, UW CSE 599H: AI vs Intelligence Augmentation**

INVITED TALKS AND INTERVIEWS

- 2026 **Cohere Labs Open Science Community**
MolmoAct2: Building Open Science Robotics Foundation Models for Real-World Deployment — Host: Cole Harrison
- 2026 **RoboPapers, Episode 87 (Interview)**
MolmoAct2: An Open Foundation for Robots That Work in the Real World
- 2026 **Talking Robotics (Interview)**
MolmoAct2: Action Reasoning Models for Real-World Deployments
- 2026 **University of Waterloo CS Seminar**
Building Robotics Foundation Models with Reasoning-in-the-Loop — Host: Victor Zhong
- 2026 **UT Austin CS Seminar**
Building Robotics Foundation Models with Reasoning-in-the-Loop — Host: Joydeep Biswas
- 2026 **University of Michigan CSE Seminar**
Building Robotics Foundation Models with Reasoning-in-the-Loop — Host: Bernadette Bucher

- 2026 **Max Planck Institute for Informatics**
Building Robotics Foundation Models with Reasoning-in-the-Loop — Host: Christian Theobalt
- 2026 **UC Irvine CS Seminar**
Building Robotics Foundation Models with Reasoning-in-the-Loop — Host: Sven Koenig
- 2025 **CoRL Workshop (Keynote)**
Grounded Reasoning from Vision-Language Models
- 2025 **TTIC Young Researcher Seminar Series**
Towards Robotics Foundation That Can Reason — Host: Matt Walter
- 2025 **Meta FAIR Robotics Group**
Towards Robotics Foundation That Can Reason — Host: Homanga Bharadhwaj
- 2025 **Mila Robot Learning Seminar**
Towards Robotics Foundation That Can Reason — Host: REAL Lab
- 2025 **Georgia Tech**
Towards Robotics Foundation That Can Reason — Host: GT Institute for Robotics and Intelligent Machines
- 2025 **TRI LBM Group**
Towards Robotics Foundation That Can Reason — Host: Jose Barreiros
- 2025 **Johns Hopkins University**
Towards Robotics Foundation That Can Reason — Host: Tianmin Shu and Peter Kazanzides (LCSR)
- 2024 **Stanford PAIR Group**
Towards Unified Multimodal LLM for Robotics — Host: Wenlong Huang
- 2024 **CMU RCHI Group**
Grounded Embodied Intelligence — Host: Zackory Erickson
- 2024 **Amazon Lab126**
Democratizing Robot Learning for All — Host: Yuyin Sun
- 2024 **The AI Talks**
Democratizing Robot Learning for All
- 2024 **UW Allen School Colloquium**
Democratizing Robot Learning for All
- 2023 **NUS CLeAR Lab**
Benchmarking Robot Learning for Manipulation — Host: Harold Soh
- 2023 **AAAI Summer Symposium**
AR2-D2: Training a Robot Without a Robot

HONORS AND AWARDS

- 2026: VLS — Outstanding Paper Award, Foundation Models Meet Embodied Agents @ CVPR; Best Paper Runner-Up, CVPR 3D-LLM/VLA Workshop.
- 2026: MolmoB0T — Best Paper Awards at three ICRA workshops: SDRL, VLA Pipeline, and Beyond Teleoperation.
- 2026: Rising Star Award for Spatial Intelligence, CVPR E2E3D Workshop — Finalist.
- 2025: SAM2Act — Best Paper Award, RememberRL @ CoRL.
- 2025: MolmoAct — Best Paper Award Runner-up, Rational Robots @ CoRL; Best Paper Award Nominee, Data @ CoRL.
- 2025: ICLR Outstanding Reviewer.
- 2024: Oral presentation at ICRA Doctoral Consortium.
- 2024: People's Choice Award, Allen School Research Showcase.
- 2024: ICLR Spotlight (Top 5%).
- 2023: Good Time to Ask — Best Paper Award, Ubiquitous Robots.
- 2021: Koh Boon Hwee Scholars Award (Top university award, 1/1392)
- 2021: National Science (PhD) Scholarship, Agency for Science, Technology and Research of Singapore (A*STAR)
- 2021: NTU Class of 1985 Book Prize
- 2021: Nanyang Award for Teamwork (Highest recognition given by the university to a student club)
- 2021: NTU EEE Excellence Award
- 2021: NTU EEE Final Year Project Challenge, 3rd Prize

- 2020: MakeMIT Hackathon, Most Innovative 5G/IoT/Sensors Design Award
- 2020: Singapore Airline APP Challenge, Superlative Award
- 2020: AI Singapore Summer School, Best Poster Award
- 2019: Singapore India Hackathon, 1st Runner-up
- 2019: 6th ASPIRE Undergraduate Research Academy Conference, Excellent Undergraduate Research Award
- 2019: Dean List's Academic Year 2018/2019
- 2019: NTU EEE Silicon Valley Study Trip Award
- 2019: A*STAR Undergraduate Scholarship
- 2016: Tan Kah Kee Young Inventor's Award (Defence Science)
- 2016: Ngee Ann Polytechnic Outstanding Technology Team Award
- 2015: Ngee Ann Polytechnic Character Award (The highest honor students with exceptional character, 1 recipient per year)

ACADEMIC SERVICE

- **Leadership:** Area Chair, CoRL 2026; Associate Editor, IEEE Robotics and Automation Letters (RA-L); Associate Editor, ICRA 2027.
- **Workshop organization:** Lead organizer for the five 2025 workshops and ERA @ CVPR 2026; organizer or co-organizer for the other workshops below.
 - 2026: From GPU-Accelerated Simulation to Scalable and Generalizable Real Robot Policy Learning @ CoRL.
 - 2026: RoboPAD: Post-Training Adaptation of Robot Foundation Models @ NeurIPS.
 - 2026: Robotic Origami Challenge @ IROS.
 - 2026: Embodied Reasoning in Action (ERA) Workshop and Challenge @ CVPR.
 - 2025: Generalization in Robotics Manipulation Workshop and Challenges @ CVPR.
 - 2025: 3D Vision Language Models (VLMs) for Robotic Manipulation: Opportunities and Challenges @ CVPR.
 - 2025: Mobile Manipulation: Emerging Opportunities & Contemporary Challenges @ RSS.
 - 2025: SPACE in Vision, Language, and Embodied AI (SpaVLE) @ NeurIPS.
 - 2025: Generalizable Priors for Robot Manipulation @ CoRL.
- **Conference reviewing:** ICRA, CVPR, ECCV, ICCV, IROS, NeurIPS, RSS, CoRL, AAAI, ICML, ICLR, and CogSci.
- **Journal reviewing:** RA-L, IEEE Transactions on Automation Science and Engineering, Pattern Recognition, Neurocomputing, and Robotics and Autonomous Systems.
- **Other service:** Chief Judge, MLDA@EEE Deep Learning Week Hackathon; General Chair, AI Student Research Conference (ARSC).

MEDIA COVERAGE

- 2026 **MolmoAct2: Action Reasoning Models for Real-World Deployment**
SiliconANGLE, TechTimes, Robotics and Automation News, The AI Economy, and The AI Insider
- 2025 **MolmoAct: Action Reasoning Models that Reason in Space**
Business Wire, The AI Economy, GeekWire, Emergent Mind, RealSense, Yahoo! Finance, and VentureBeat
- 2024 **AR2-D2: Training a Robot Without a Robot**
MIT Technology Review

REFERENCES

- **Prof. Dieter Fox**
Professor, Paul G. Allen School of Computer Science & Engineering, University of Washington
fox@cs.washington.edu — Ph.D. Advisor
- **Prof. Ranjay Krishna**
Assistant Professor, Paul G. Allen School of Computer Science & Engineering, University of Washington
ranjay@cs.washington.edu — Ph.D. Advisor and Collaborator
- **Prof. Ali Farhadi**
CEO, Allen Institute for AI (AI2); Professor, University of Washington
afarhad2@gmail.com — Research Collaborator and Mentor